

2(a)(i)	T_1 and T_2 : down T_3 : up $\left. \vphantom{\begin{matrix} T_1 \\ T_2 \\ T_3 \end{matrix}} \right\}$ All 3 must be correct.	B1
2(a)(ii)	By Principle of Moments, pivot at base of wire 1, sum of clockwise moments = sum of anticlockwise moments $(14.5)W = (10.0)T_3$ $T_3 = \frac{14.5}{10.0}(m_p g)$ $= \frac{14.5}{10.0}(350 \times 10^{-3})(9.81)$ $= 4.98 \text{ N}$ $\left. \vphantom{\begin{matrix} \text{pivot at base of wire 1,} \\ \text{sum of clockwise moments = sum of anticlockwise moments} \end{matrix}} \right\}$	M1 A1

Qns	Answer	Marks
3(a)	limit for which light reflecting off the 2 features can be just	